**HW3 Assessment Criteria**

| | Excellent | Good | Satisfactory | Unsatisfactory |
|---|---|---|---|---|
| **Functionality (20 pt.)** | The synthesized design based on the previous two assignments passes more than 90% of the test cases. **(18-20 pt.)** | The synthesized design based on the previous two assignments passes more than 75% of the test cases. **(15-17 pt.)** | The synthesized design based on the previous two assignments passes more than 50% of the test cases. **(10-14 pt.)** | The synthesized design based on the previous two assignments passes less than 50% of the test cases. **(0-9 pt.)** |
| **Synthesis Flow (25 pt.)** | Provide a detailed explanation of the synthesis software workflow with accompanying diagrams. Besides, use additional synthesis commands **(24-25 pt.)** | Provide a detailed explanation of the synthesis software workflow with accompanying diagrams. **(18-23 pt.)** | Provide a brief explanation of the synthesis software workflow, but the explanation is somewhat lacking. **(13-17 pt.)** | Provide a concise overview of the synthesis software workflow, but there are errors in the operations or results, and they have not been addressed. **(0-12 pt.)** |
| **Circuit Analysis (15 pt.)** | Use tools to analyze the post-synthesis logic gate count, logic gate levels, annotated longest paths, power consumption, etc., clearly. Provide diagrams illustrating these aspects, and make comparisons with the previous pre-synthesis homework. **(13-15 pt.)** | Use tools to analyze the post-synthesis logic gate count, logic gate levels, annotated longest paths, power consumption, etc., clearly. **(11-13 pt.)** | Use tools to analyze the post-synthesis logic gate count, logic gate levels, annotated longest paths, power consumption, etc., but not clear enough. **(7-10 pt.)** | Use tools to analyze the post-synthesis logic gate count, logic gate levels, annotated longest paths, power consumption, etc., but the result is wrong. **(0-6 pt.)** |
| **Hardware Performance (5 pt)** | Hardware performance is in the top 20% of the class. **(5 pt.)** | Hardware performance is in the top 60% of the class. **(4-3 pt.)** | Hardware performance is in the top 90% of the class. **(1-2 pt.)** | Hardware performance is below 90% of the class. **(0 pt.)** |

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| **Testbench and completeness of verification (15 pt.)** | The verification and waveforms are clear and complete, considering various corner cases, based on HW1 and HW2. The verification signal generation and comparison in testbench are performed automatically. **(14-15 pt.)** | The verification and waveforms are clear and complete, considering various corner cases. However, the verification signal generation and comparison in testbench are performed manually. **(11-13 pt.)** | Complete the basic verification, perform the waveform. However only consider part of corner cases. **(7-10 pt.)** | Only partial verification and wave pattern display are performed. The completeness is not good. **(0-6 pt.)** |
| **Discussion and Reflection (15 pt.)** | Clearly explain the problems and solutions encountered during the synthesis process. And reflect on the impact on the tool usage in the future. **(14-15 pt.)** | Clearly explain the problems encountered during the synthesis process and their solutions. But there is no reflection. **(11-13 pt.)** | It only describes some of the problems encountered during the synthesis process and solutions. Not complete enough. **(7-10 pt.)** | Only briefly state the problems encountered during the synthesis process and solutions. Poor structure. **(0-6 pt.)** |
| **Report Overall (5 pt.)** | Reports are clear and easy to read. Fonts, font styles, line spacing, indentations, pictures, etc. are neat and unified, and the paragraph hierarchy is clear and distinct. **(5 pt.)** | Reports are clear and easy to read. Fonts, font styles, line spacing, indentations, pictures, etc. are neat and unified, However, the paragraph hierarchy is no clear enough. **(3-4 pt.)** | Only some fonts, font styles, line spacing, indentations, pictures, etc. of the report are neat. The paragraph hierarchy is not complete enough. **(1-2 pt.)** | The fonts, font styles, line spacing, indentations, pictures and other formats in the report are disordered, and the paragraph hierarchical structure is poor. **(0 pt.)** |